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1. How many unique values does **Company** have? **Job Title**?

Answer 1: Company have 12 unique values and 9 job title

SS1:

The screenshot shows the SAS Visual Analytics interface. On the left, the 'Data' pane lists the 'EMPLOYEES' table. A tooltip is displayed over the 'Company' column, showing the following information:

- Name: Company
- Distinct values: 12
- Name in data: Company
- Format: \$
- Used by: ListTable.1 role

The main table displays the following columns: Employee Hire Date, Employee Country, Employee Name, Employee Termination Date, Group, Job Title, and Section. The data is as follows:

Employee Hire Date	Employee Country	Employee Name	Employee Termination Date	Group	Job Title	Section
January 1, 1978		Sian Shannan	June 30, 2010	Internet/Catalog Sales Management	Sales Rep. II	Sales
October 1, 2010		Petrea Soltau		Golf	Sales Rep. II	Sales
January 1, 1978		Caterina Hayawardhana		Racket Sports	Sales Rep. III	Sales
September 1, 2010		Fang Wilson		Shoes	Sales Rep. II	Sales
November 1, 2010		Koavea Pa		Team Sports	Sales Rep. I	Sales
December 1, 2010		Leonid Karavdic		Racket Sports	Sales Rep. II	Sales
January 1, 1997		Judy Chantharasy		Outdoors	Sales Rep. I	Sales
January 1, 2002		Samantha Waal	January 31, 2010	Racket Sports	Sales Rep. I	Sales
July 1, 1982		Christina Ngan		Assorted Sports Articles	Sales Rep. II	Sales
January 1, 1978		Fadi Nowd	August 31, 2010	Running - Jogging	Sales Rep. IV	Sales
October 1, 2001		Alexei Platts	April 30, 2008	Indoor Sports	Sales Rep. IV	Sales
March 1, 2010		Gerry Snellings		Racket Sports	Sales Rep. I	Sales
October 1, 1982		Fancine Kaiser		Golf	Sales Rep. III	Sales
February 1, 2002		Franca Kierce		Team Sports	Sales Rep. III	Sales
March 1, 2010		Marina Iyengar		Shoes	Sales Rep. III	Sales
January 1, 2007		Kerrin Dillin	June 30, 2007	Trainees	Trainee	Sales
April 1, 2009		Sean Dives		Racket Sports	Sales Rep. I	Sales
October 1, 2010		Michael Minas		Outdoors	Sales Rep. I	Sales

2. What is the type (or classification) of **Employee ID**?

Answer 2: Measure is the classification of Employee ID

SS2:

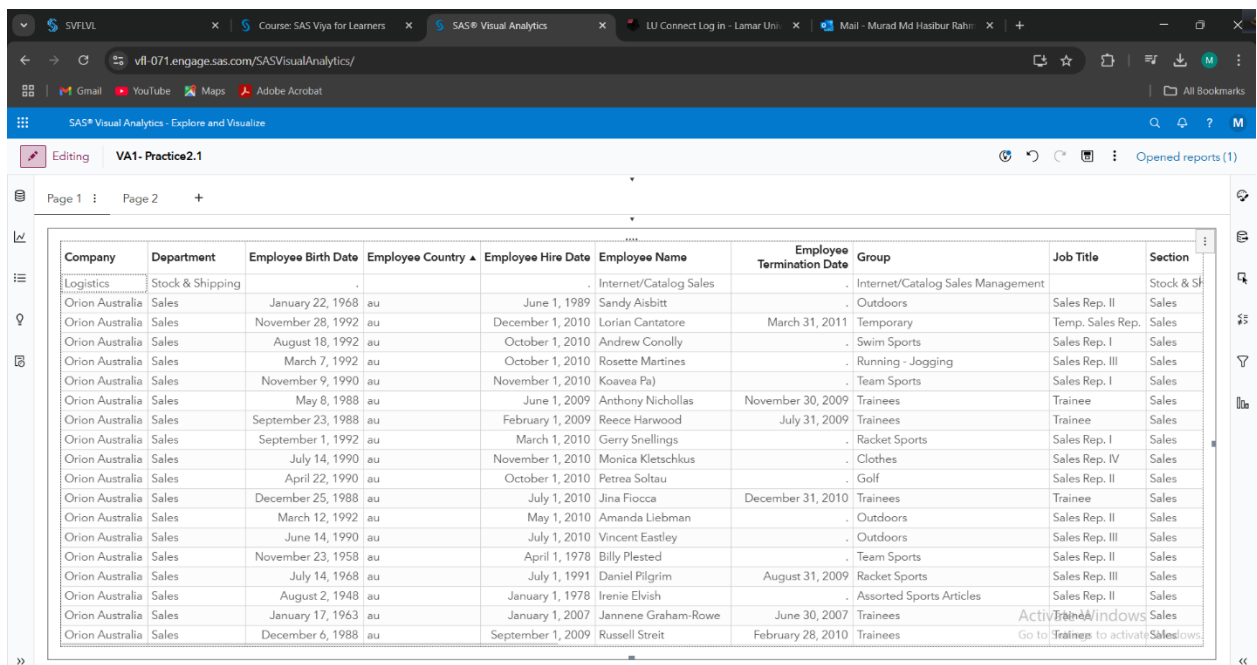
The screenshot displays the SAS Visual Analytics interface. The main window shows a data table with columns: Birth Date, Employee Country, Employee Hire Date, Employee Name, Employee Termination Date, Group, Job Title, and Section. The table contains 20 rows of employee data. On the left sidebar, the 'Data' section is active, showing a list of measures. The 'Employee ID' measure is selected, and its classification is set to 'Measure'. The format is set to 'Float (F12.)' and the aggregation is set to 'Default (Sum)'. The 'Job Title' measure is also visible in the list. The interface includes a top navigation bar with tabs for 'Course: SAS Viya for Learners', 'SAS Visual Analytics', and 'LU Connect Log in - Lamar Uni'. The bottom status bar shows 'Opened reports (1)'.

Birth Date	Employee Country	Employee Hire Date	Employee Name	Employee Termination Date	Group	Job Title	Section	
.	.	.	Internet/Catalog Sales	.	Internet/Catalog Sales Management	.	Stock & k	
6, 1953	au	January 1, 1978	Sian Shannan	June 30, 2010	Golf	Sales Rep. II	Sales	
2, 1990	au	October 1, 2010	Petrea Soltau	.	Golf	Sales Rep. II	Sales	
0, 1948	au	January 1, 1978	Caterina Haywardhana	.	Racket Sports	Sales Rep. III	Sales	
8, 1990	au	September 1, 2010	Fang Wilson	.	Shoes	Sales Rep. II	Sales	
9, 1990	au	November 1, 2010	Koavea Pa	.	Team Sports	Sales Rep. I	Sales	
1, 1990	au	December 1, 2010	Leonid Karavdic	.	Racket Sports	Sales Rep. II	Sales	
1, 1978	au	January 1, 1997	Judy Chantharasy	.	Outdoors	Sales Rep. I	Sales	
7, 1983	au	January 1, 2002	Samantha Waal	January 31, 2010	Racket Sports	Sales Rep. I	Sales	
7, 1958	au	July 1, 1982	Christina Ngen	.	Assorted Sports Articles	Sales Rep. II	Sales	
4, 1948	au	January 1, 1978	Fadi Nowd	August 31, 2010	Running - Jogging	Sales Rep. IV	Sales	
6, 1973	au	October 1, 2001	Alexei Platts	April 30, 2008	Indoor Sports	Sales Rep. IV	Sales	
1, 1992	au	March 1, 2010	Gerry Snellings	.	Racket Sports	Sales Rep. I	Sales	
5, 1953	au	October 1, 1982	Fancine Kaiser	.	Golf	Sales Rep. III	Sales	
5, 1983	au	February 1, 2002	Franca Kierce	.	Team Sports	Sales Rep. III	Sales	
2, 1983	au	March 1, 2010	Marina Iyengar	.	Shoes	Sales Rep. III	Sales	
8, 1958	au	January 1, 2007	Kerrin Dillin	June 30, 2007	Trainees	Trainee	Sales	
1, 1983	au	April 1, 2009	Sean Dives	.	Racket Sports	Sales Rep. I	Sales	
3, 1992	au	October 1, 2010	Michael Minas	.	Outdoors	Sales Rep. I	Sales	
.	.	.	October 1, 1982	Fancine Kaiser	.	Golf	Sales Rep. III	Sales
.	.	.	February 1, 2002	Franca Kierce	.	Team Sports	Sales Rep. III	Sales
.	.	.	March 1, 2010	Marina Iyengar	.	Shoes	Sales Rep. III	Sales
.	.	.	January 1, 2007	Kerrin Dillin	June 30, 2007	Trainees	Trainee	Sales
.	.	.	April 1, 2009	Sean Dives	.	Racket Sports	Sales Rep. I	Sales
.	.	.	October 1, 2010	Michael Minas	.	Outdoors	Sales Rep. I	Sales

3. What is the case of **Employee Country (Upper case /Lower case)?**

Answer 3: The case of Employee Country is Lower case

SS3:



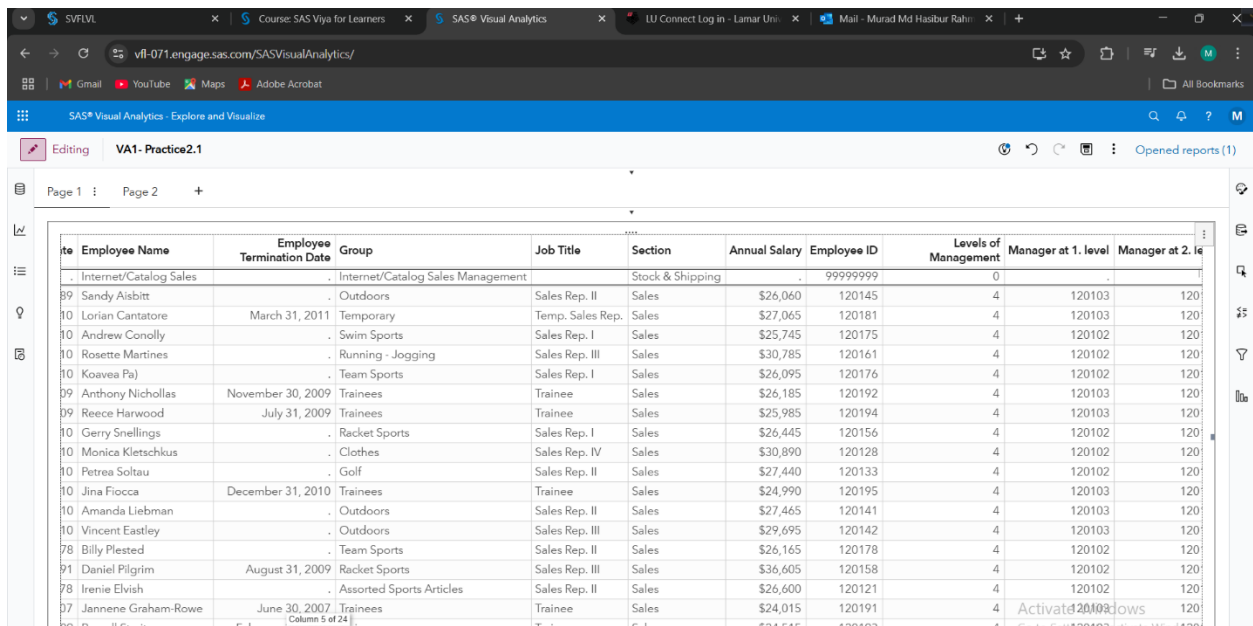
The screenshot displays the SAS Visual Analytics interface. The browser address bar shows the URL `vfl-071.engage.sas.com/SASVisualAnalytics/`. The interface includes a top navigation bar with the text "SAS® Visual Analytics - Explore and Visualize". Below this, a tab labeled "VA1- Practice2.1" is active. The main content area shows a data table with the following columns: Company, Department, Employee Birth Date, Employee Country, Employee Hire Date, Employee Name, Employee Termination Date, Group, Job Title, and Section. The table contains 20 rows of employee data, including details for various departments like Logistics, Sales, and Trainees, and job titles such as Sales Rep. I, Sales Rep. II, and Sales Rep. III. The interface also features a sidebar on the left with navigation icons and a right sidebar with additional options. The bottom of the interface shows a status bar with the text "Active windows" and "Go to: Trainings to activate Sales".

Company	Department	Employee Birth Date	Employee Country	Employee Hire Date	Employee Name	Employee Termination Date	Group	Job Title	Section
Logistics	Stock & Shipping	.	.	.	Internet/Catalog Sales	.	Internet/Catalog Sales Management	Stock & Shipping	.
Orion Australia	Sales	January 22, 1968	au	June 1, 1989	Sandy Aisbitt	.	Outdoors	Sales Rep. II	Sales
Orion Australia	Sales	November 28, 1992	au	December 1, 2010	Lorian Cantatore	March 31, 2011	Temporary	Temp. Sales Rep.	Sales
Orion Australia	Sales	August 18, 1992	au	October 1, 2010	Andrew Conolly	.	Swim Sports	Sales Rep. I	Sales
Orion Australia	Sales	March 7, 1992	au	October 1, 2010	Rosette Martines	.	Running - Jogging	Sales Rep. III	Sales
Orion Australia	Sales	November 9, 1990	au	November 1, 2010	Koavea Pa	.	Team Sports	Sales Rep. I	Sales
Orion Australia	Sales	May 8, 1988	au	June 1, 2009	Anthony Nichollas	November 30, 2009	Trainees	Trainee	Sales
Orion Australia	Sales	September 23, 1988	au	February 1, 2009	Reece Harwood	July 31, 2009	Trainees	Trainee	Sales
Orion Australia	Sales	September 1, 1992	au	March 1, 2010	Gerry Snellings	.	Racket Sports	Sales Rep. I	Sales
Orion Australia	Sales	July 14, 1990	au	November 1, 2010	Monica Kletschkus	.	Clothes	Sales Rep. IV	Sales
Orion Australia	Sales	April 22, 1990	au	October 1, 2010	Petrea Soltau	.	Golf	Sales Rep. II	Sales
Orion Australia	Sales	December 25, 1988	au	July 1, 2010	Jina Fiocca	December 31, 2010	Trainees	Trainee	Sales
Orion Australia	Sales	March 12, 1992	au	May 1, 2010	Amanda Liebman	.	Outdoors	Sales Rep. II	Sales
Orion Australia	Sales	June 14, 1990	au	July 1, 2010	Vincent Eastley	.	Outdoors	Sales Rep. III	Sales
Orion Australia	Sales	November 23, 1958	au	April 1, 1978	Billy Plested	.	Team Sports	Sales Rep. II	Sales
Orion Australia	Sales	July 14, 1968	au	July 1, 1991	Daniel Pilgrim	August 31, 2009	Racket Sports	Sales Rep. III	Sales
Orion Australia	Sales	August 2, 1948	au	January 1, 1978	Irenie Elvish	.	Assorted Sports Articles	Sales Rep. II	Sales
Orion Australia	Sales	January 17, 1963	au	January 1, 2007	Jannene Graham-Rowe	June 30, 2007	Trainees	Trainee	Sales
Orion Australia	Sales	December 6, 1988	au	September 1, 2009	Russell Streit	February 28, 2010	Trainees	Trainee	Sales

4.Which employee represents sales over the internet or through the catalog?

Answer 4: Internet/Catalog sales Management represents sales over the internet or through the catalog.

SS4:



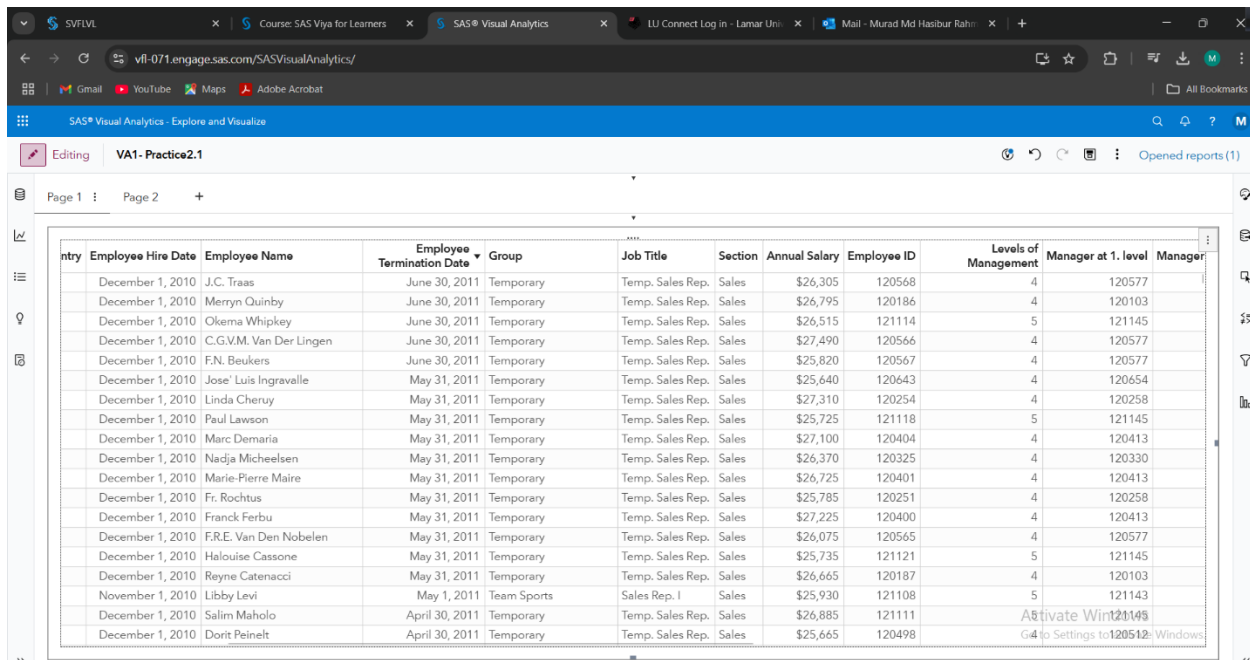
The screenshot shows the SAS Visual Analytics interface. The browser address bar displays 'vfl-071.engage.sas.com/SASVisualAnalytics/'. The application title is 'SAS® Visual Analytics - Explore and Visualize'. The report is titled 'VA1- Practice2.1'. The table below lists employee data with columns for Employee Name, Termination Date, Group, Job Title, Section, Annual Salary, Employee ID, Levels of Management, Manager at 1. level, and Manager at 2. level.

Employee Name	Employee Termination Date	Group	Job Title	Section	Annual Salary	Employee ID	Levels of Management	Manager at 1. level	Manager at 2. level
Internet/Catalog Sales		Internet/Catalog Sales Management		Stock & Shipping		99999999	0		
Sandy Aisbitt		Outdoors	Sales Rep. II	Sales	\$26,060	120145	4	120103	120103
Lorian Cantatore	March 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$27,065	120181	4	120103	120103
Andrew Conolly		Swim Sports	Sales Rep. I	Sales	\$25,745	120175	4	120102	120102
Rosette Martines		Running - Jogging	Sales Rep. III	Sales	\$30,785	120161	4	120102	120102
Koavea Pa		Team Sports	Sales Rep. I	Sales	\$26,095	120176	4	120102	120102
Anthony Nichollas	November 30, 2009	Trainees	Trainee	Sales	\$26,185	120192	4	120103	120103
Reece Harwood	July 31, 2009	Trainees	Trainee	Sales	\$25,985	120194	4	120103	120103
Gerry Snellings		Racket Sports	Sales Rep. I	Sales	\$26,445	120156	4	120102	120102
Monica Kletschkius		Clothes	Sales Rep. IV	Sales	\$30,890	120128	4	120102	120102
Petrea Soltau		Golf	Sales Rep. II	Sales	\$27,440	120133	4	120102	120102
Jina Fiocca	December 31, 2010	Trainees	Trainee	Sales	\$24,990	120195	4	120103	120103
Amanda Liebman		Outdoors	Sales Rep. II	Sales	\$27,465	120141	4	120103	120103
Vincent Eastley		Outdoors	Sales Rep. III	Sales	\$29,695	120142	4	120103	120103
Billy Plested		Team Sports	Sales Rep. II	Sales	\$26,165	120178	4	120102	120102
Daniel Pilgrim	August 31, 2009	Racket Sports	Sales Rep. III	Sales	\$36,605	120158	4	120102	120102
Irenie Elvish		Assorted Sports Articles	Sales Rep. II	Sales	\$26,600	120121	4	120102	120102
Jannene Graham-Rowe	June 30, 2007	Trainees	Trainee	Sales	\$24,015	120191	4	120103	120103
Bruce E. Smith		Trainees	Trainee	Sales	\$24,515	120193	4	120103	120103

5. Which data item can be used to determine whether an employee is active (currently employed) or retired (formerly employed)?

Answer 5: Employee Hire date and Employee Termination Date be used to determine whether an employee is active or retired .

SS5:



The screenshot shows the SAS Visual Analytics interface with a table titled 'VA1 - Practice2.1'. The table contains employee data with columns for Hire Date, Employee Name, Termination Date, Group, Job Title, Section, Annual Salary, Employee ID, Levels of Management, Manager at 1. level, and Manager at 2. level. The data is sorted by Hire Date in descending order.

Country	Employee Hire Date	Employee Name	Employee Termination Date	Group	Job Title	Section	Annual Salary	Employee ID	Levels of Management	Manager at 1. level	Manager at 2. level
	December 1, 2010	J.C. Traas	June 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,305	120568	4	120577	
	December 1, 2010	Merryn Quinby	June 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,795	120186	4	120103	
	December 1, 2010	Okerna Whipkey	June 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,515	121114	5	121145	
	December 1, 2010	C.G.V.M. Van Der Lingen	June 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$27,490	120566	4	120577	
	December 1, 2010	F.N. Beukers	June 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$25,820	120567	4	120577	
	December 1, 2010	Jose Luis Ingravalles	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$25,640	120643	4	120654	
	December 1, 2010	Linda Cheruy	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$27,310	120254	4	120258	
	December 1, 2010	Paul Lawson	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$25,725	121118	5	121145	
	December 1, 2010	Marc Demaria	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$27,100	120404	4	120413	
	December 1, 2010	Nadja Micheelsen	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,370	120325	4	120330	
	December 1, 2010	Marie-Pierre Maire	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,725	120401	4	120413	
	December 1, 2010	Fr. Rochtus	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$25,785	120251	4	120258	
	December 1, 2010	Franck Ferbu	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$27,225	120400	4	120413	
	December 1, 2010	F.R.E. Van Den Nobelen	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,075	120565	4	120577	
	December 1, 2010	Halouise Cassone	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$25,735	121121	5	121145	
	December 1, 2010	Reyne Catenacci	May 31, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,665	120187	4	120103	
	November 1, 2010	Libby Levi	May 1, 2011	Team Sports	Sales Rep. I	Sales	\$25,930	121108	5	121143	
	December 1, 2010	Salim Maholo	April 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$26,885	121111	4	120512	
	December 1, 2010	Dorit Peinelt	April 30, 2011	Temporary	Temp. Sales Rep.	Sales	\$25,665	120498	4	120512	

6. Which department contains the missing job title?

Answer 6: Stock & Shipping department contains the missing job title.

SS6:

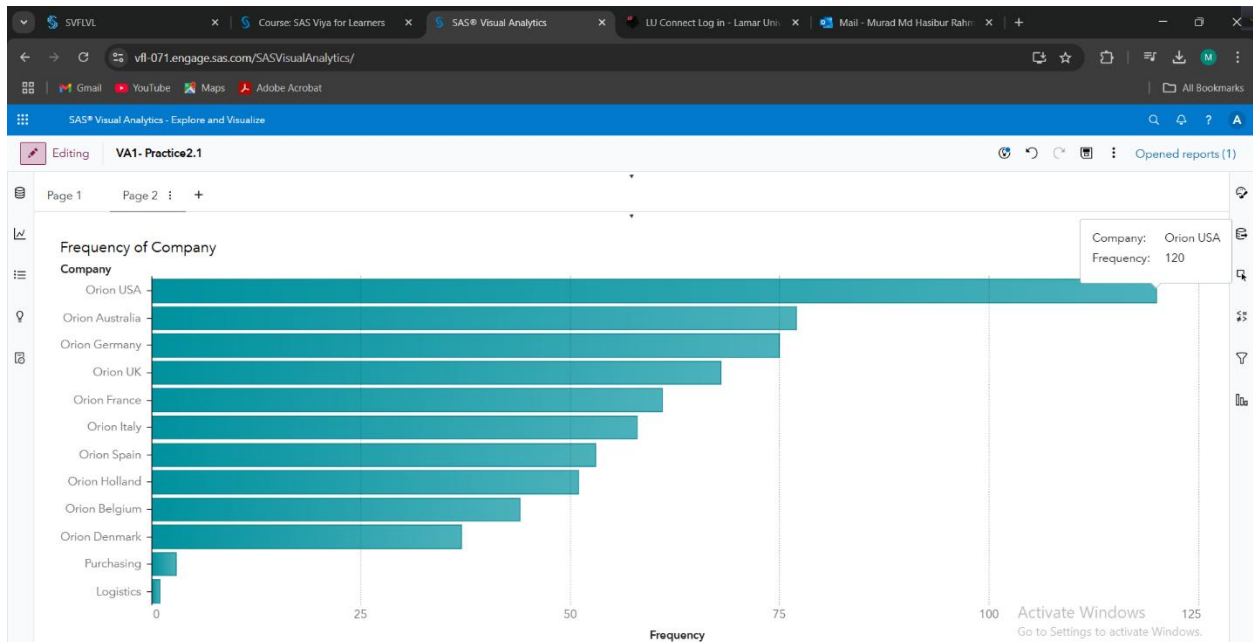
The screenshot shows the SAS Visual Analytics interface with a table titled 'VA1-Practice2.1'. The table has three columns: Department, Job Title, and Frequency. The data is as follows:

Department	Job Title	Frequency
Purchasing	Purchasing Agent I	1
	Purchasing Agent III	2
Sales	Sales Rep. I	215
	Sales Rep. II	172
	Sales Rep. III	113
	Sales Rep. IV	49
	Temp. Sales Rep.	70
	Trainee	25
Stock & Shipping	(missing)	1

7. What is the largest company, based on the number of employees? The smallest?

Answer 7: The largest company, based on the number of employees is Orion USA, Frequency :120 and the lowest is Logistics, Frequency :1

SS7:



8. What is the minimum total profit generated by an employee? The maximum? The average? The total profit generated by all employees?

Answer 8: The minimum total profit generated by an employee is \$11.10, The maximum is \$19146779.62 and The average is \$109k

SS8:

SAS® Visual Analytics - Explore and Visualize

Editing VA1- Practice2.1

Page 1 Page 2

Manager at 1. level	Manager at 2. level	Manager at 3. level	Manager at 4. level	Manager at 5. level	Manager at 6. level	Total Customers	Total Orders	Total Products Ordered	Total Profit ▲	Total Quantity Ordered
121145	121142	121141	120261	120259	.	1	1	1	\$11.10	1
120512	120510	120261	120259	.	.	4	4	4	\$102.95	4
120512	120510	120261	120259	.	.	5	5	5	\$108.60	5
120258	120257	120261	120259	.	.	7	7	9	\$195.60	9
120577	120576	120261	120259	.	.	7	7	11	\$225.05	11
120258	120257	120261	120259	.	.	8	8	11	\$232.15	11
120103	120101	120261	120259	.	.	3	3	4	\$243.60	10
121143	121142	121141	120261	120259	.	12	12	15	\$283.45	15
120330	120331	120261	120259	.	.	10	10	17	\$309.35	21
120258	120257	120261	120259	.	.	3	3	3	\$386.25	5
120512	120510	120261	120259	.	.	4	4	5	\$389.70	10
120989	120988	120261	120259	.	.	8	8	10	\$418.80	23
120654	120653	120261	120259	.	.	15	15	19	\$432.20	19
120577	120576	120261	120259	.	.	3	3	3	\$535.50	9
120881	120880	120261	120259	.	.	18	18	24	\$543.05	24
120102	120101	120261	120259	.	.	4	4	5	\$586.30	10
120511	120510	120261	120259	.	.	19	19	30	\$593.40	30
Sum:						87	87	87	\$70,727,947.65	87

SAS Visual Analytics - Explore and Visualize

EditingVA1-Practice2.1

Opened reports (1)

Page 1Page 2

Manager at 1. level	Manager at 2. level	Manager at 3. level	Manager at 4. level	Manager at 5. level	Manager at 6. level	Total Customers	Total Orders	Total Products Ordered	Total Profit	Total Quantity Ordered
-	-	-	-	-	-	187,420	186,636	235,699	\$19,146,779.62	412,684
120415	120412	120261	120259	-	-	5,628	5,625	7,136	\$395,552.03	11,857
120415	120412	120261	120259	-	-	5,603	5,593	7,064	\$390,366.78	11,635
121144	121142	121141	120261	120259	-	5,448	5,446	6,904	\$377,979.02	11,536
121144	121142	121141	120261	120259	-	5,329	5,327	6,795	\$374,494.31	11,347
121144	121142	121141	120261	120259	-	5,457	5,456	6,848	\$372,456.18	11,329
121144	121142	121141	120261	120259	-	5,438	5,437	6,920	\$372,296.83	11,349
121144	121142	121141	120261	120259	-	5,438	5,437	6,857	\$371,596.96	11,399
121144	121142	121141	120261	120259	-	5,347	5,345	6,843	\$371,506.09	11,315
120881	120880	120261	120259	-	-	5,365	5,364	6,797	\$366,162.87	11,205
121144	121142	121141	120261	120259	-	5,211	5,211	6,621	\$363,717.62	10,939
120577	120576	120261	120259	-	-	2,795	2,793	3,558	\$333,371.34	5,832
120511	120510	120261	120259	-	-	3,359	3,359	4,082	\$332,087.50	6,918
120511	120510	120261	120259	-	-	4,646	4,644	5,942	\$332,004.77	9,863
120511	120510	120261	120259	-	-	3,258	3,257	4,032	\$329,628.32	6,829
120511	120510	120261	120259	-	-	4,615	4,614	5,937	\$327,051.90	9,771
120577	120576	120261	120259	-	-	2,774	2,770	3,504	\$325,707.85	5,714
Sum:						2,700	2,700	4,244	\$70,727,947.65	4,046

Explaining Total Profit

Total Profit ranges from \$11 to \$19M. Average Total Profit is \$109K. Most cases (the middle 80%) have a Total Profit between \$2.3K and \$255K. Total Customers best differentiates the highest (top 10%) and the lowest (bottom 10%) Total Profit cases. The three most related factors are Job Title, Manager at 2. level, and Employee Country.

There are 51 cases that might be outliers, with Total Profit greater than or equal to \$266K.

Add full explanation to new pageClose

SAS Visual Analytics - Explore and Visualize

EditingVA1-Practice2.1

Opened reports (1)

Page 1Page 2

Manager at 1. level	Manager at 2. level	Manager at 3. level	Manager at 4. level	Manager at 5. level	Manager at 6. level	Total Customers	Total Orders	Total Products Ordered	Total Profit	Total Quantity Ordered
-	-	-	-	-	-	187,420	186,636	235,699	\$19,146,779.62	412,684
120415	120412	120261	120259	-	-	5,628	5,625	7,136	\$395,552.03	11,857
120415	120412	120261	120259	-	-	5,603	5,593	7,064	\$390,366.78	11,635
121144	121142	121141	120261	120259	-	5,448	5,446	6,904	\$377,979.02	11,536
121144	121142	121141	120261	120259	-	5,329	5,327	6,795	\$374,494.31	11,347
121144	121142	121141	120261	120259	-	5,457	5,456	6,848	\$372,456.18	11,329
121144	121142	121141	120261	120259	-	5,438	5,437	6,920	\$372,296.83	11,349
121144	121142	121141	120261	120259	-	5,438	5,437	6,857	\$371,596.96	11,399
121144	121142	121141	120261	120259	-	5,347	5,345	6,843	\$371,506.09	11,315
120881	120880	120261	120259	-	-	5,365	5,364	6,797	\$366,162.87	11,205
121144	121142	121141	120261	120259	-	5,211	5,211	6,621	\$363,717.62	10,939
120577	120576	120261	120259	-	-	2,795	2,793	3,558	\$333,371.34	5,832
120511	120510	120261	120259	-	-	3,359	3,359	4,082	\$332,087.50	6,918
120511	120510	120261	120259	-	-	4,646	4,644	5,942	\$332,004.77	9,863
120511	120510	120261	120259	-	-	3,258	3,257	4,032	\$329,628.32	6,829
120511	120510	120261	120259	-	-	4,615	4,614	5,937	\$327,051.90	9,771
120577	120576	120261	120259	-	-	2,774	2,770	3,504	\$325,707.85	5,714
Sum:						2,700	2,700	4,244	\$70,727,947.65	4,046

Explaining Total Profit

Total Profit ranges from \$11 to \$19M. Average Total Profit is \$109K. Most cases (the middle 80%) have a Total Profit between \$2.3K and \$255K. Total Customers best differentiates the highest (top 10%) and the lowest (bottom 10%) Total Profit cases. The three most related factors are Job Title, Manager at 2. level, and Employee Country.

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Add full explanation to new pageClose